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[22AECME51]

FIFTH SEMESTER B. TECH EXAMINATION, DECEMBER-2024

CONDITION MONITORING AND MAINTENANCE MANAGEMENT

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A			**5 x2=10**
1.	**a.**	Write the functions of maintenance	L1	CO1	**02**
	b.	Define Run-out Maintenance.	L1	CO2	**02**
	c.	List the goals of TPM.	L1	CO3	**02**
	d.	Define fishbone diagram.	L1	CO4	**02**
	e.	Define digital signal processing.	L1	CO5	**02**
		PART-B			**6 X 5=30**
2.	**a.**	Explain spare parts management system.	L2	CO1	**05**
		OR			
	b.	Discuss on maintenance policy goals.	L2	CO1	**05**
3.	**a.**	Discuss probability theory a with examples in maintenance management.	L2	CO2	**05**
		OR			
	b.	Write short on RCM.	L2	CO2	**05**
4.	**a.**	Explain Pareto analysis.	L2	CO3	**05**
		OR			
	b.	Explain key performance indicators of TPM.	L2	CO3	**05**
5.	**a.**	Prepare the preventative maintenance checklist for two wheeler.	L3	CO4	**05**
		OR			
	b.	Explain subjective condition monitoring and objective condition monitoring.	L2	CO4	**05**
6.	**a.**	Discuss fault diagnosis for rotational machines.	L3	CO5	**05**
		OR			
	b.	Prepare a FMEA chart for mobile phones.	L3	CO5	**05**
7.	**a.**	Discuss on the development of maintenance system.	L2	CO1	**05**
		OR			

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	b.	Explain MTTR and MTBF.	L2	CO2	**05**
		PART-C	**6 X 10=60**		
8.	**a.**	Discuss predictive maintenance with example.	L2	CO1	**10**
		OR			
	b.	List different types of maintenance strategies and explain any one with examples	L2	CO1	**10**
9.	**a.**	Explain bathtub curve for electric blub.	L2	CO2	**10**
		OR			
	b.	Write the procedure for conducting RCM analysis	L2	CO2	**10**
10.	**a.**	Interpret 5's for the food package industries.	L3	CO3	**10**
		OR			
	b.	Sketch and explain fishbone diagram for cellular phone.	L3	CO3	**10**
11.	**a.**	Explain the condition monitoring process for food package industries.	L3	CO3	**10**
		OR			
	b.	Explain the levels of condition monitoring with example.	L3	CO3	**10**
12.	**a.**	Write different methods of Condition monitoring techniques and any one with neat sketch	L3	CO5	**10**
		OR			
	b.	Explain digital signal processing and computational techniques in maintenance system.	L3	CO5	**10**
13.	**a.**	Explain 8 pillars of TPM.	L5	CO6	**10**
		OR			
	b.	Explain **i.** Failure probability **ii.** Redundancy **iii.** Hazard rate function **iv.** MTTR **v.** MTBF	L5	CO6	**10**